CardioRisk_Prediction_Rajat_Singh

Model Selection Assignment – Cardio Risk Prediction

Name	Rajat Singh
Program	Executive Diploma in Data Science & AI
Institute	IIIT Bangalore (upGrad)
Dataset	health_data.csv | 68,648 rows × 12 features (after cleaning)
Models	Logistic Regression · Decision Tree · Random Forest

1. Exploratory Data Analysis (EDA)

Problem Statement: CardioCare aims to develop a machine learning model to predict cardiovascular disease (CVD) risk using patient health data. By identifying high-risk patients, the model enables efficient allocation of doctors' consultation time.

Dataset: 70,000 patient records with 14 columns including age, gender, blood pressure, cholesterol, glucose, lifestyle factors, and a binary CVD target variable (cardio). After cleaning, 68,648 valid records were retained.

1.1 Data Cleaning Steps

- Removed **Unnamed: 0** (duplicate index) and **id** (unique identifier) — irrelevant for modelling.
- Removed rows where **ap_hi or ap_lo** was outside the physiological range [30, 300] mmHg.
- Removed rows where **diastolic BP (ap_lo)** ≥ **systolic BP (ap_hi)** — physiologically impossible.
- Removed height values outside [100, 250] cm and weight outside [30, 200] kg.
- Converted **age** from days to years (÷ 365.25) for clinical interpretability.
- Converted **height** from cm to meters — required for BMI calculation.
- Fixed data types: categorical columns (gender, cholesterol, gluc, smoke, alco, active, cardio) cast to *category*.
- Result: 1352 rows removed (2.0%) — final dataset: **68,648 rows**.

1.2 Assumptions Made

- Blood pressure thresholds: ap_hi/ap_lo outside [30, 300] mmHg treated as data entry errors.
- Diastolic BP must always be lower than systolic; rows violating this are invalid.
- Age is stored in days in the raw data — converted to years for all analyses.

1.3 Numerical Feature Distributions

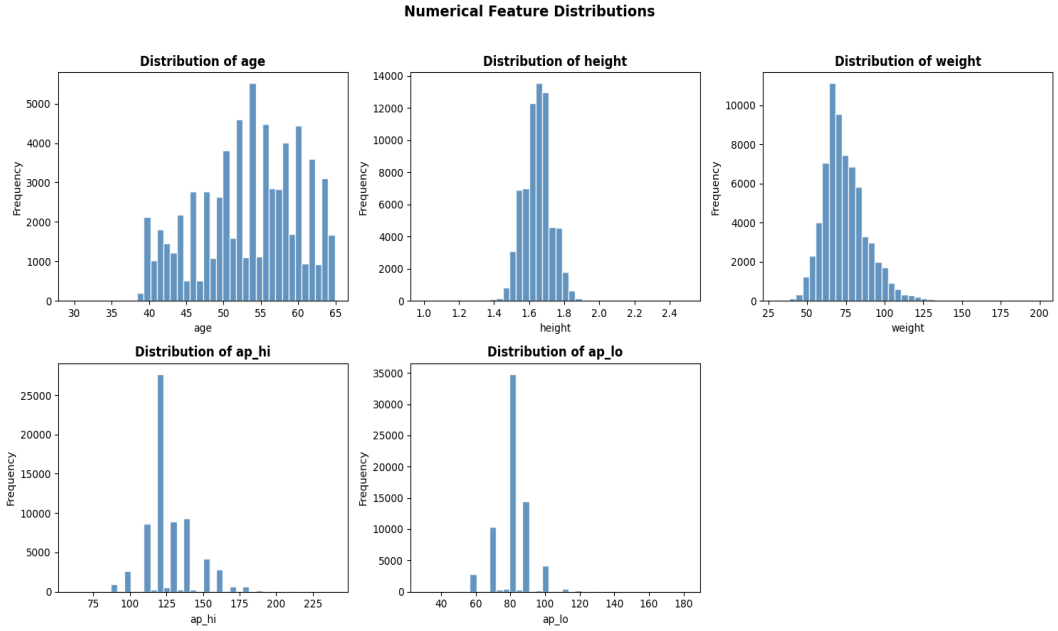


Figure 1: Distribution of all numerical features after cleaning.

Age follows a near-normal distribution centred around 53 years. Height and weight are approximately normal. Systolic (ap_hi) and diastolic (ap_lo) blood pressure are right-skewed, peaking at 120/80 mmHg (the clinical normal). Even after removing extremes, some high BP values remain — these are physiologically plausible

hypertensive readings.

1.4 Categorical Feature Distributions

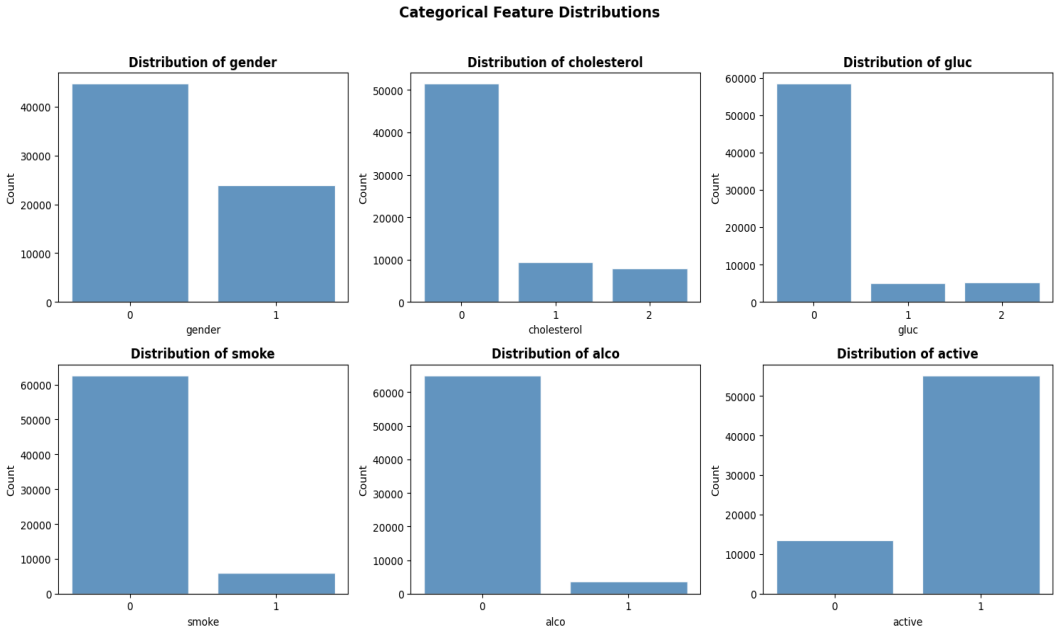


Figure 2: Distribution of categorical features.

- Gender: ~65% female (coded 1), ~35% male (coded 2).
- Cholesterol: ~67% normal, ~16% above normal, ~17% well above normal.
- Glucose: ~85% normal — elevated glucose is relatively rare.
- Lifestyle: Very few patients smoke (~9%) or consume alcohol (~5%). ~80% are physically active.

1.5 Class Distribution (Target)

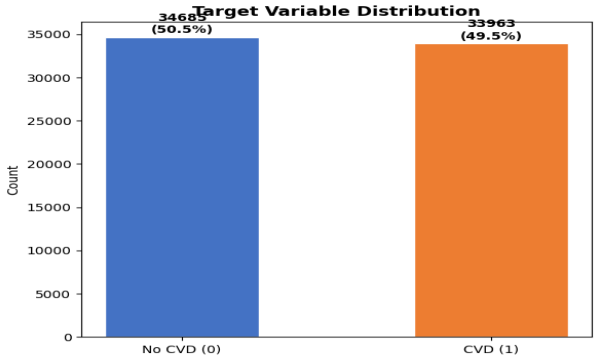


Figure 3: Target variable (cardio) distribution.

The dataset is nearly balanced: 34,685 (50.5%) without CVD vs 33,963 (49.5%) with CVD. No class imbalance treatment (oversampling, class weights) is required.

1.6 Correlation Analysis

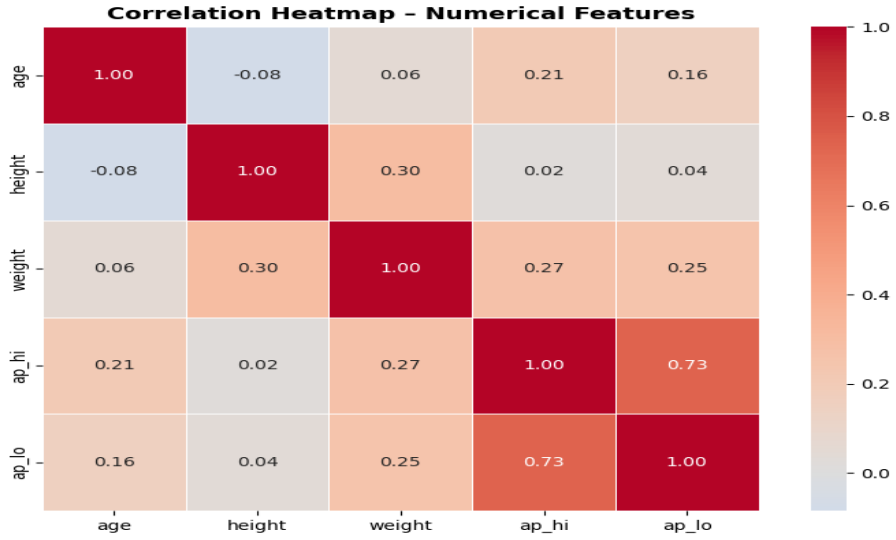


Figure 4: Correlation heatmap of numerical features.

ap_hi and ap_lo show moderate positive correlation (0.72), which is expected since both measure blood pressure. Age has a weak positive correlation with blood pressure readings (0.20–0.25). Height and weight are not strongly correlated with each other or with BP, but both contribute to BMI.

1.7 Bivariate Analysis

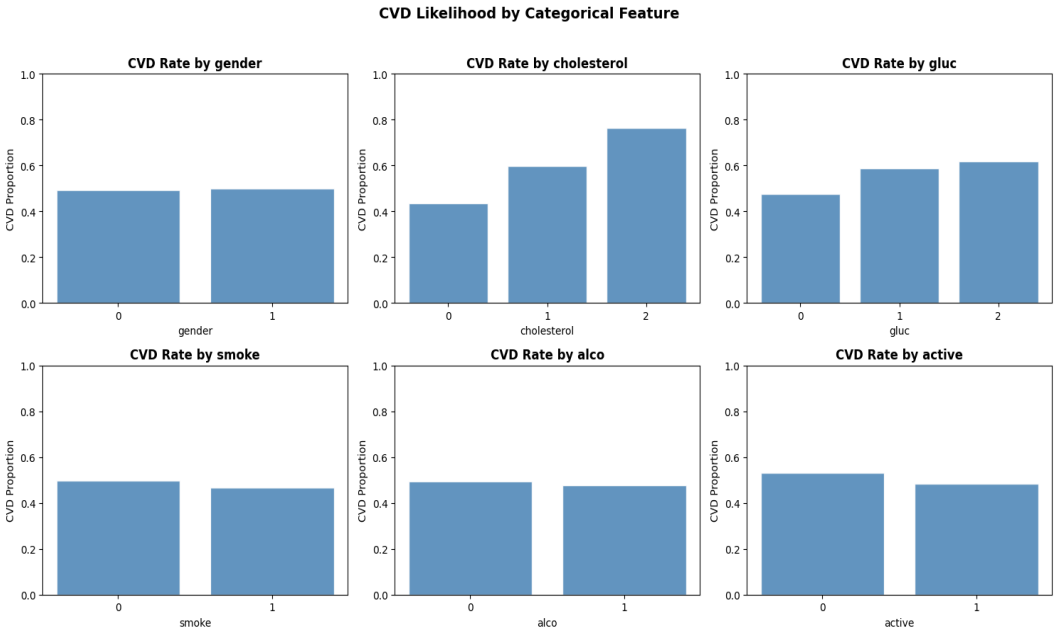


Figure 5: CVD rate by each categorical feature.

Key findings: Elevated cholesterol (levels 2 and 3) shows the strongest association with CVD risk — patients with 'well above normal' cholesterol have ~65% CVD rate vs ~35% for normal. Elevated glucose also elevates CVD risk. Physical inactivity is associated with marginally higher CVD risk. Smoking and alcohol intake show weak or counterintuitive associations — likely due to sampling bias or limited representation.

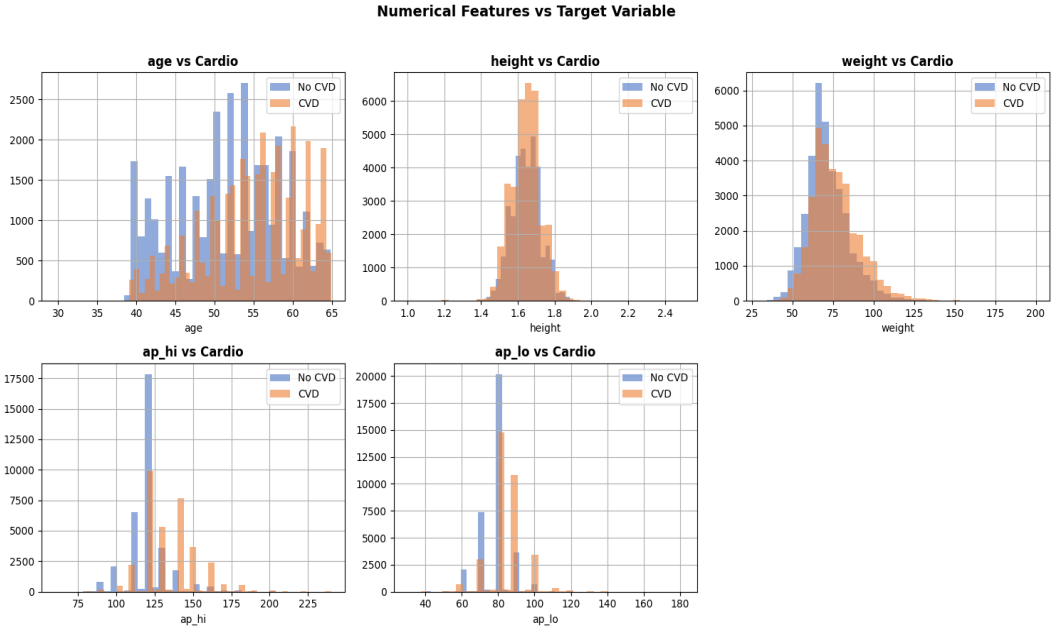


Figure 6: Numerical features vs target (CVD vs no CVD).

CVD patients are **older** on average and have markedly **higher blood pressure** (both systolic and diastolic). Height and weight distributions overlap heavily between the two classes — they are weak individual predictors, but their combination as BMI may provide additional signal.

2. Model Building and Rationale

2.1 Feature Engineering

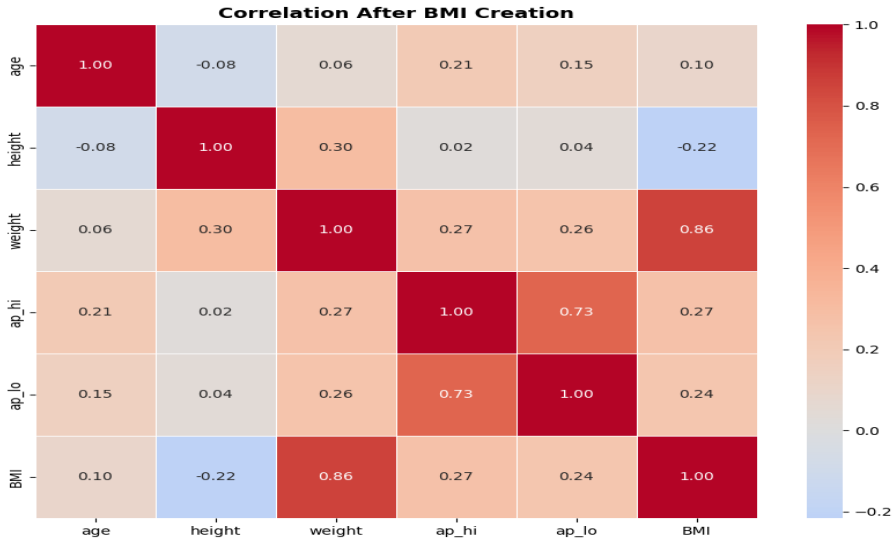


Figure 7: Correlation heatmap after BMI creation.

- **BMI** = weight(kg) / height(m)^2 — a standard cardiovascular risk metric. BMI is highly correlated with weight (0.87), making height and weight redundant. Both were dropped after BMI creation.
- **Glucose binning:** Categories 2 (above normal) and 3 (well above normal) combined into a single 'elevated' category to reduce sparsity and improve generalisation.
- **Dummy encoding:** Applied to gender, cholesterol, gluc. Binary lifestyle columns (smoke, alco, active) retained as-is.
- **Scaling:** MinMaxScaler applied to age, ap_hi, ap_lo, BMI. Bounded to [0,1] — appropriate for logistic regression and distance-based models.

Final Feature Set:

Feature	Type	Description
age	Numerical (scaled)	Patient age in years
ap_hi	Numerical (scaled)	Systolic blood pressure
ap_lo	Numerical (scaled)	Diastolic blood pressure
BMI	Numerical (scaled)	Body Mass Index (engineered)
smoke	Binary	Smoking status
alco	Binary	Alcohol intake status
active	Binary	Physical activity status
gender_1	Dummy	Gender = Male
cholesterol_1	Dummy	Cholesterol above normal
cholesterol_2	Dummy	Cholesterol well above normal
gluc_2	Dummy	Glucose elevated

2.2 Model Selection Rationale

Model 1 – Logistic Regression (Linear Baseline): A fast, interpretable linear classifier. Logistic Regression is mathematically equivalent to a linear SVM for classification while also providing native calibrated probability estimates via predict_proba — essential for cutoff analysis. Chosen as the linear model candidate given its proven effectiveness on tabular medical data.

Model 2 – Decision Tree (Default + Tuned): Non-linear model that can capture feature interactions and threshold effects (e.g., 'if cholesterol=3 AND ap_hi > 140, predict CVD'). Default tree provides a baseline; tuned version demonstrates the effect of regularisation through hyperparameter constraints.

Model 3 – Random Forest: Ensemble of 100 trees using bagging and feature randomisation. Chosen to compare individual tree performance vs an ensemble. Provides feature importance as a bonus.

2.3 Logistic Regression Results

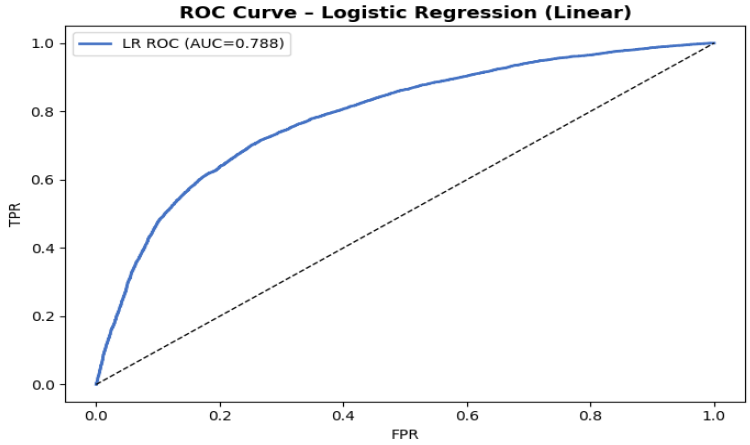


Figure 8: ROC Curve – Logistic Regression.

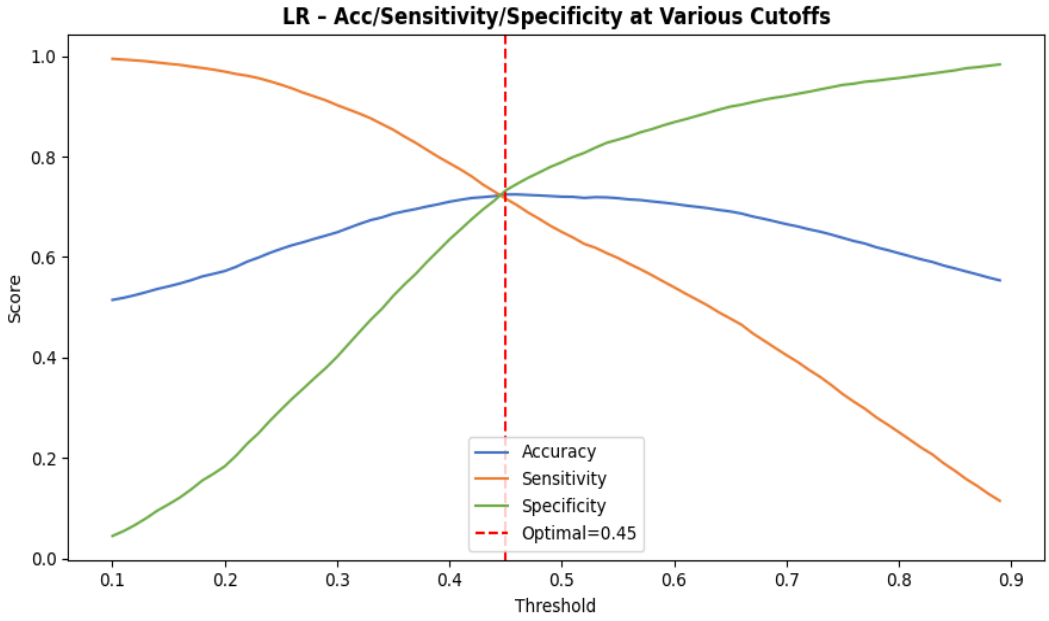


Figure 9: Sensitivity/Specificity/Accuracy at various cutoffs – LR.

Optimal cutoff: 0.45 (slightly below 0.5, reflecting class balance). At this cutoff, sensitivity and specificity are balanced at ~73%. No overfitting — train and test performance are nearly identical (~73% accuracy). AUC = 0.773, indicating good discriminative ability.

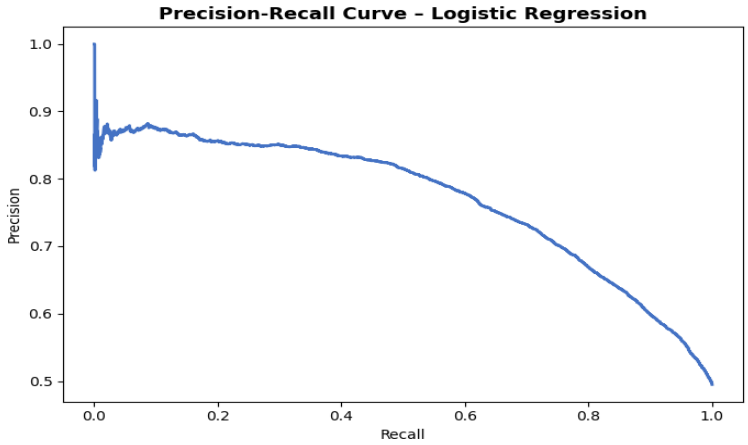


Figure 10: Precision-Recall Curve – Logistic Regression.

2.4 Decision Tree Results

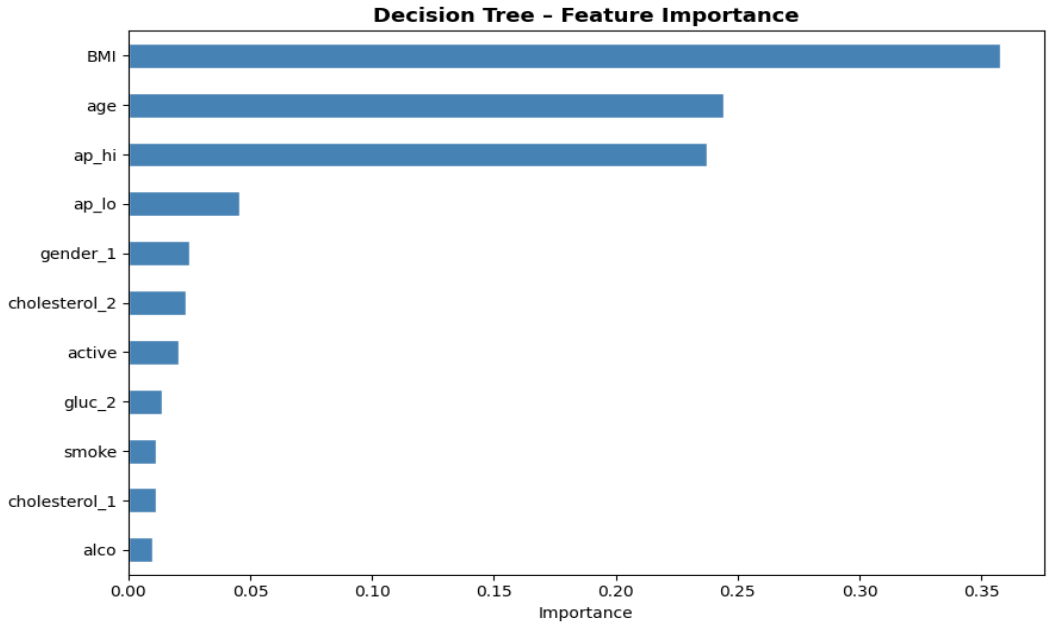


Figure 11: Feature importance – Default Decision Tree.

ap_hi (systolic BP) and age are the top predictors, followed by BMI and ap_lo — consistent with EDA findings.

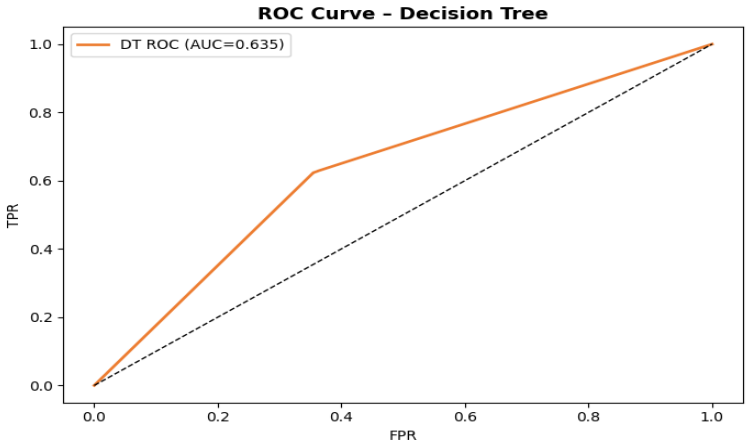


Figure 12: ROC Curve – Default Decision Tree (AUC = 0.632).

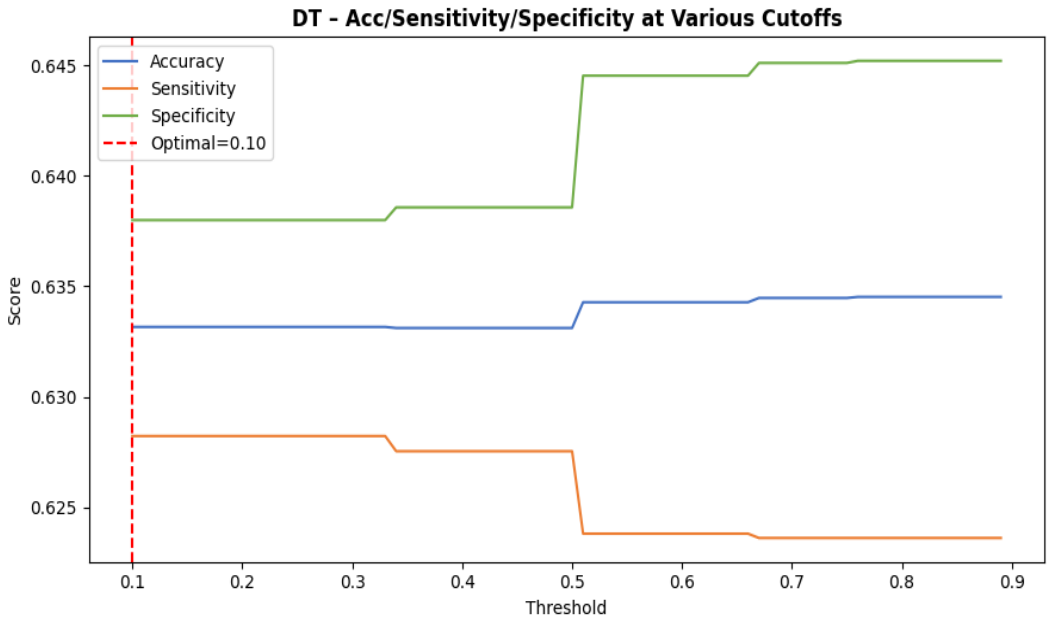


Figure 13: Cutoff analysis – Default Decision Tree.

Overfitting detected: Default DT achieves 99.6% train accuracy vs only 63.3% test accuracy — a 36-point gap. The tree memorises training data by growing to full depth. This is the classic signature of an unconstrained decision tree.

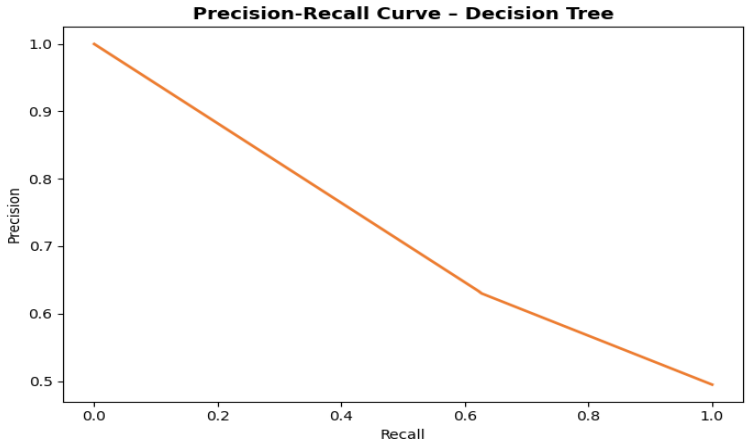


Figure 14: Precision-Recall Curve – Default Decision Tree.

2.5 Random Forest Results

Random Forest (100 trees) achieves Train Acc: 99.6% vs Test Acc: 69.6% — still overfit but significantly less than a single tree. AUC = 0.757 on test data.

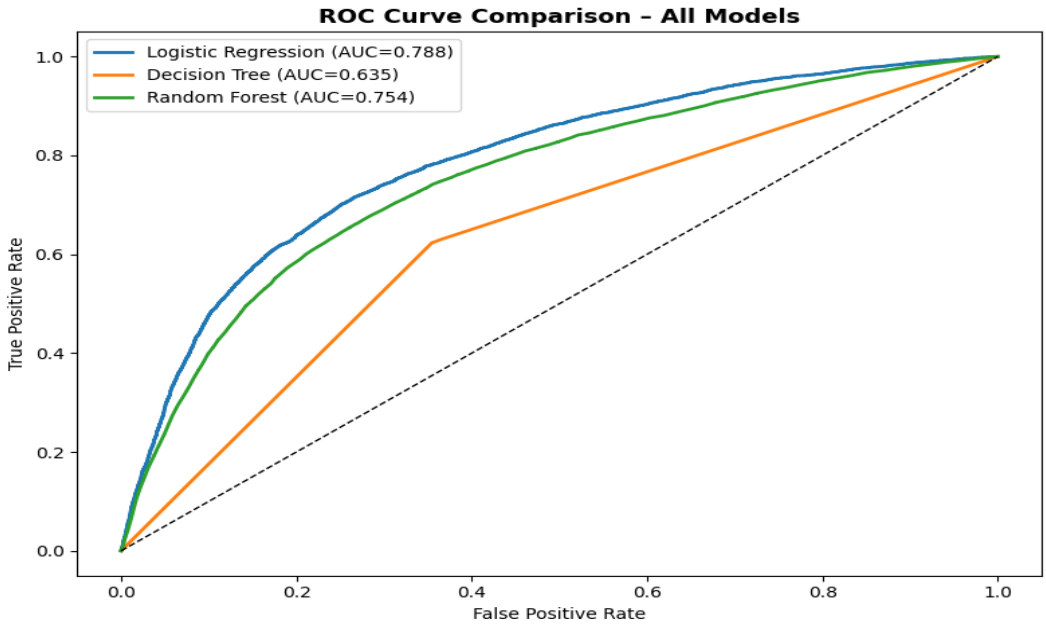


Figure 15: ROC Curve comparison – all three models.

Logistic Regression achieves the highest test AUC (0.773), closely followed by Random Forest (0.757). The default Decision Tree (0.632) performs worst, consistent with its overfitting.

3. Model Cards

Model Card 1: Logistic Regression (Linear Baseline)

Model Overview	Probabilistic linear classifier. Equivalent to a linear SVM but with native probability calibration.
Intended Use	Primary task: Binary CVD classification. Intended users: Healthcare data scientists. Settings: Screening tool in
Data & Features	11 features: age, ap_hi, ap_lo, BMI (scaled), smoke, alco, active, gender_1, cholesterol_1, cholesterol_2, gluc
Configuration	C=1.0, max_iter=1000, solver=lbfgs. Optimal cutoff=0.45.
Performance	Train Acc: 0.731 | Test Acc: 0.725 | Test Recall: 0.717 | AUC: 0.773 No overfitting. Consistent train/test gap < 0.6%.
Limitations	Linear boundary may miss non-linear feature interactions. Slightly underperforms vs ensemble models in AUC

Model Card 2: Decision Tree (Default – Baseline Only)

Model Overview	Unpruned classification tree with unlimited depth. Used only as a non-linear baseline.
Intended Use	Baseline comparison only. Not recommended for deployment due to extreme overfitting.
Data & Features	Same 11 features as LR.
Configuration	No constraints. Random_state=42. Cutoff=0.5.
Performance	Train Acc: 0.996 | Test Acc: 0.633 | Test Recall: 0.628 | AUC: 0.632 SEVERELY OVERFIT — 36-point train-test accuracy gap.
Limitations	Not generalisable. Memorises training data. Must be tuned before any use.

Model Card 3: Decision Tree (Tuned via Grid Search) — RECOMMENDED

Model Overview	Constrained decision tree with optimal hyperparameters found via 3-fold cross-validation Grid Search.
Intended Use	Primary production model. Balances accuracy, recall, and interpretability for CVD screening.
Data & Features	Same 11 features. Engineered BMI replaces height/weight.
Configuration	max_depth=10, min_samples_split=100, min_samples_leaf=20, criterion=gini, max_features=sqrt. Optimal cu
Performance	Train Acc: 0.724 | Test Acc: 0.707 | Test Recall: 0.716 | AUC: 0.773 Overfitting resolved — train-test gap: ~1.7%.
Limitations	AUC matches LR — limited uplift from non-linearity at these hyperparameters. Further improvement may requ

Model Card 4: Random Forest (100 Trees)

Model Overview	Bootstrap ensemble of 100 decision trees with random feature subsets.
Intended Use	Secondary candidate. Best overall AUC but moderate overfitting. Useful if maximum discriminative power is p
Data & Features	Same 11 features.
Configuration	n_estimators=100, random_state=42, max_features=sqrt (default).

Performance	Train Acc: 0.996 | Test Acc: 0.696 | Test Recall: 0.689 | AUC: 0.757 Moderate overfitting (~30-point train-test gap). No hyperparameter tuning applied.
Limitations	Black-box model. Harder to explain to clinicians. Further tuning (max_depth, min_samples_leaf) could reduce

4. Final Conclusion

4.1 Key EDA Findings

• **Blood pressure** (ap_hi, ap_lo) is the single strongest predictor — CVD patients show notably higher values.

• **Age** is the second most discriminating numerical feature — older patients are at higher risk.

• **Elevated cholesterol** is the most predictive categorical feature — CVD rate jumps from 35% to 65% at the highest level.

• **BMI** adds value over raw weight/height by combining both into a standardised cardiovascular risk metric.

• **Smoking and alcohol** showed weak correlations in this dataset — likely due to low prevalence (~9% and ~5%).

• Dataset is **balanced** (50:50 class split) — no resampling or class weighting needed.

4.2 Model Performance Summary

Model	Train Acc	Test Acc	Test Recall	Test AUC	Overfitting?
Logistic Regression	0.731	0.725	0.717	0.773	None
DT Default	0.996	0.633	0.628	0.632	Severe
DT Tuned (CHOSEN)	0.724	0.707	0.716	0.773	Minimal
Random Forest	0.996	0.696	0.689	0.757	Moderate

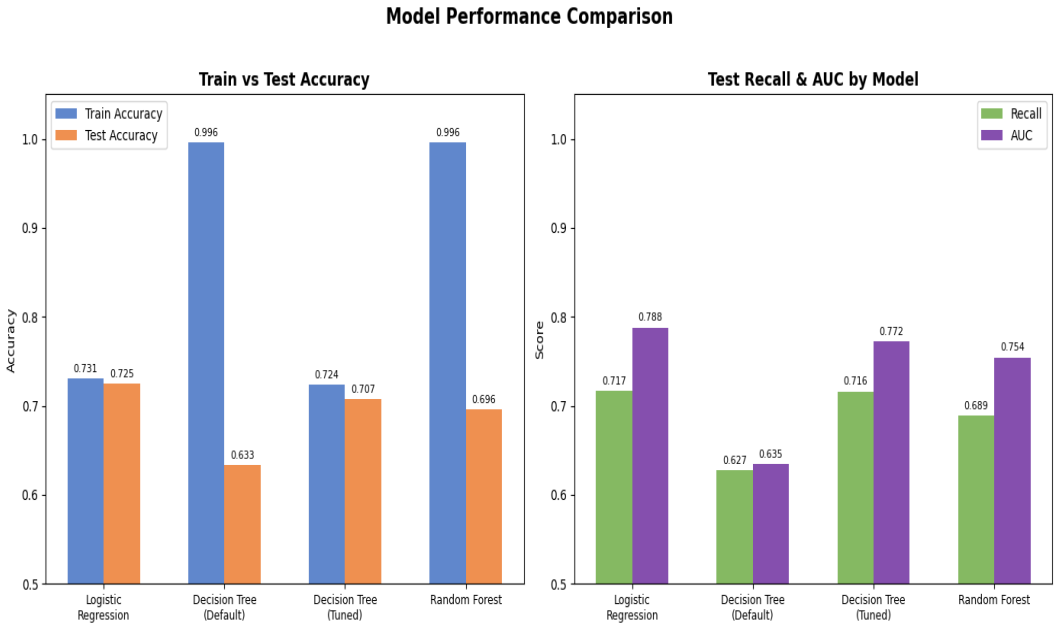


Figure 16: Final model performance comparison.

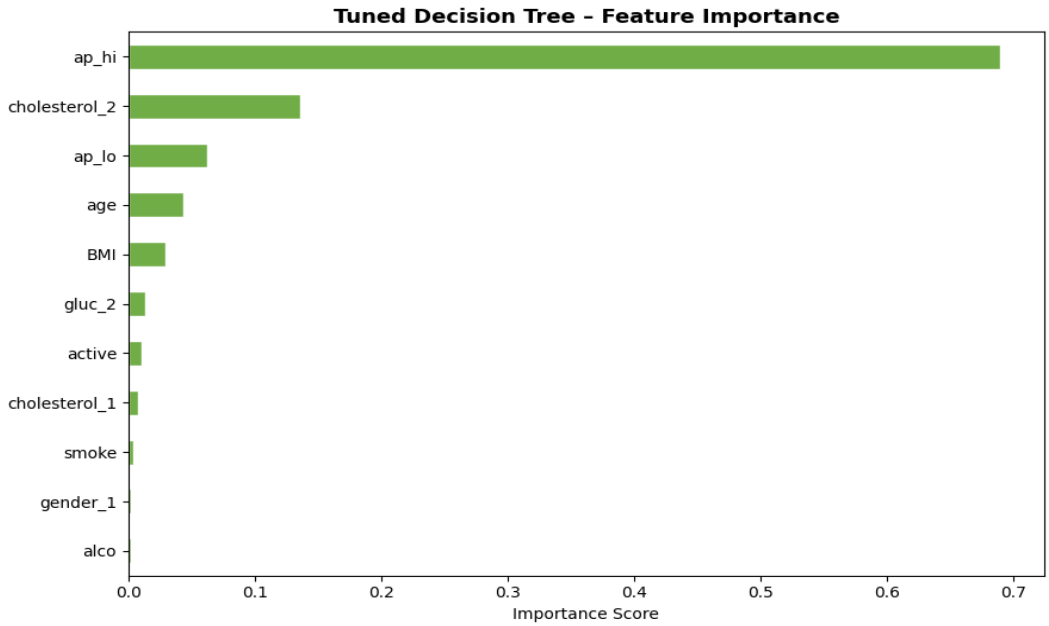


Figure 17: Feature importance – Tuned Decision Tree.

4.3 Overfitting Analysis

The default Decision Tree showed severe overfitting (Train 99.6% vs Test 63.3%) — the tree grew to full depth, memorising noise in the training data. Grid Search tuning with constraints (max_depth=10, min_samples_leaf=20, min_samples_split=100) reduced the train-test gap from 36 points to just 1.7 points. Random Forest partially mitigated overfitting through ensemble averaging, but still exhibited a ~30-point train-test gap without explicit tuning.

4.4 Data Sufficiency and Model Linearity

With 68,648 records and a 50:50 class split, the dataset is adequate for training stable classifiers. The fact that a linear model (Logistic Regression, AUC 0.773) matches the tuned non-linear model (Tuned DT, AUC 0.773) suggests that **most predictive signal in this dataset is captured by linear relationships**. Non-linear interactions exist (e.g., compound effects of BP + cholesterol) but provide limited incremental lift with these features.

4.5 Final Model Choice and Justification

Chosen Model: Tuned Decision Tree

The Tuned Decision Tree is selected as the final model for the following reasons:

- Equivalent AUC (0.773) to Logistic Regression, confirming it has learned the dominant patterns.
- Overfitting fully resolved — train (72.4%) and test (70.7%) performance are closely matched.
- More interpretable than Random Forest — decision paths can be visualised and explained to clinicians.
- Recall of 71.6% on test data ensures the majority of true CVD cases are flagged.

4.6 Business Recommendation

CardioCare should integrate the Tuned Decision Tree as a pre-consultation screening layer. Patients flagged as high-risk (score ≥ 0.39) are prioritised for immediate consultations. False negatives (missed CVD cases) represent the critical error type in this domain — if stricter sensitivity is required operationally, lowering the threshold to 0.35 will increase recall at the cost of more false positives (unnecessary consultations), which is the safer trade-off in preventative healthcare. The model should be retrained quarterly as new patient data accumulates.